

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all elements above the leading entry position must be 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ is in reduced row echelon form.

~~The~~ The 3rd row has no pivot position.

By Theorem 4, if A is an $m \times n$ matrix, then A has a pivot position in every row if and only if for each b in $\mathbb{R}^m$ the equation $Ax=b$ has a solution.

$\therefore$ ~~A~~ A has at least one row with no pivot position if and only if there exists a b in $\mathbb{R}^m$ s.t. the equation $Ax=b$ has no solution.

$\therefore$ Since the matrix $A = \begin{bmatrix} a & b \\ c & d \\ \text{etc} & \text{etc} \end{bmatrix}$ does not have a pivot position in every row, the equation $Ax=b$ cannot have a solution for some b , i.e. it is inconsistent. $\square$

Let A be an arbitrary $m \times n$ matrix. Let, $m < n$. From Theorem 1, we know matrix A has a unique reduced echelon form. By defn, a pivot position of a matrix corresponds to a leading 1 of one of its rows in reduced row echelon form. A reduced row echelon form of a matrix must have all non-zero rows above the zero rows; for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it, and for each column having a leading entry, all the elements below the leading entry position must be 0.

From this condns, we conclude that if all the n columns have pivots, then the last $(m-n)$ rows are zero rows. $\therefore$ All rows do not have pivots

From Theorem 4, we know that for an arbitrary $m \times n$ matrix A , for each b in $\mathbb{R}^m$, $Ax=b$ has a solution if and only if A has a pivot in every row. Since A does not have a pivot for every row, there exists some b in $\mathbb{R}^m$ for which $Ax=b$ has no solution, i.e. it is inconsistent. $\square$